



AgentFuse

A Logical Circuit Breaker for Long-Range AI Agents

Autonomous agents fail quietly — infinite loops, goal drift, logic traps, runaway spend. AgentFuse is an independent supervisor that catches these in real time and steers the agent back on track.

Observability

Safety / Guardrails

OpenAI AgentKit

TEAM

Brocode

ns-0437.github.io/agentfuse

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Introduction

What exactly is AgentFuse?

Think of the circuit breaker in your home — when current spikes, it trips and cuts the power before damage is done, then a human resets it.

AgentFuse is that breaker, for AI agents.

Modern agents run for hours across hundreds of steps with no human watching. AgentFuse is an independent supervisor: it watches every action, trips when the agent goes wrong, and uses a separate model to steer it back — or safely escalates to a human.



Watches everything

Tool calls, graph routes, state and token spend



Detects failure

Loops, goal drift, logic stalls, runaway spend



Steers back

A separate model writes a fix, injected live



Knows its limits

Escalates to a human when it can't be fixed safely

Problem Statement

Long-range autonomy fails quietly — and the agent can't see it happening.



Infinite Tool Loops

Repeats the same action over and over, expecting a different result.



Goal Drift

Slowly forgets the objective and wanders off onto an unrelated task.



Logic Traps

Reasons flawlessly — but from a false assumption it never questions.



Runaway Spend

Burns tokens (real money) step after step with nothing to show for it.



The catch: *the agent that failed is the worst judge of its own failure — you need an independent supervisor.*

The Solution — How It Works

1

Watch

Observe every tool call, route and token in real time

2

Detect

Four sensors: loops · drift · stalls · runaway spend

3

Trip

Freeze the run the moment a danger threshold is crossed

4

Steer

A separate reasoning model writes the fix, injected back

5

Resume / Escalate

Continue on the corrected path — or hand to a human

LIVE DEMO

```
agentfuse · logical circuit breaker

$ python examples/demo_loop_trap.py

OBJECTIVE Rotate the production database credential.

step 1 tool_call search_files(/config, *.conn)
step 1 tool_result 0 files matched
step 2 tool_call search_files(/config, *.conn)
step 2 tool_result 0 files matched
step 3 tool_call search_files(/config, *.conn)

CIRCUIT BREAKER TRIPPED - LOOP
search_files called 3x, identical args, no state progress.

STEERING RECOVERY · separate reasoning model
Stop repeating search_files - it isn't advancing the task.
Re-read the objective and take a different next action.

resume steering injected - agent resuming
step 4 tool_call secret_manager.get(prod/db/primary)
COMPLETE credential rotated after self-healing

trips 1 recoveries 1 steps 4 status complete
```

A real agent stuck in a loop — caught by the breaker and steered back to finish, in one run.

Tech Stack

Framework-agnostic core, first-class OpenAI AgentKit integration.



Python

Stdlib-only core — runs anywhere



OpenAI AgentKit

First-class, real RunHooks



OpenAI SDK

Guarded tool-use loop



LangGraph

Callback-handler adapter



Reasoning Model

Separate-model recovery (o4-mini)



JSONL Traces

Full run observability



HTML / JS Dashboard

Self-contained, no server



GitHub Pages

Free live deployment

Live Demo & Dashboard

⚡ **AgentFuse**

Logical circuit breaker · run observability

Goal Drift complete

Agent slowly wanders off-objective; the breaker re-anchors it to the original goal.

⚡ 1 trips · 🔄 1 recoveries · 4500 tok

Human Escalation escalated

An unrecoverable failure; the breaker hard-stops and hands control to a human.

⚡ 3 trips · 🔄 3 recoveries · 17100 tok

Infinite Tool Loop complete

Agent repeats a doomed tool call; the breaker detects it and forces a new path.

⚡ 1 trips · 🔄 1 recoveries · 3750 tok

Real AgentKit Run complete

Live openai-agents Runner + real hooks; a real run that self-healed.

⚡ 1 trips · 🔄 1 recoveries · 5400 tok

Goal Drift

Summarize the Q3 revenue figures from the attached finance report into three bullet points.

recovery backend: mock · loop23 · drift<0.45 · max_tokens - · trace: runs/drift.jsonl

STATUS
complete

STEPS
4

TOKENS
4,500

COST
\$0.018

TRIPS
1

RECOVERIES
1

CUMULATIVE TOKEN SPEND



peak cumulative: 4,500 tokens

DETECTORS

loop drift ⚡ progress spend

GRAPH ROUTE

analyst → supervisor → analyst

EXECUTION TIMELINE

step 1 llm_call Opening the finance report to read the Q3 revenue section.

step 2 llm_call Q3 revenue looks tied to the new product line's marketing push.

⚡ **CIRCUIT BREAKER TRIPPED — DRIFT · trip**

Agent's working goal has drifted from the original objective (similarity 0.28 < 0.45 for 2 consecutive turns).

🔄 **STEERING RECOVERY** inject

Why: Interpreted goal diverged from the system prompt; re-anchor to original objective. (conf 0.75, via mock)

You have drifted from your assigned objective. Your recent focus ("Q3 revenue looks tied to the new product line's marketing push.") is not what you were asked to do. Discard that tangent and realign strictly to the ORIGINAL objective: "Summarize the Q3 revenue figures from the attached finance report into three bullet points.". State your next action in terms of that objective only.

step 2 resume ▶ re-anchored to original objective

step 3 llm_call Re-focusing: extracting the three key Q3 revenue figures from the report.

step 4 complete ✓ Q3 revenue summarized in three bullets

LIVE

See every intervention

Every supervised run, replayable — trips, steering recoveries, token spend and graph routes at a glance.

- Timeline of trips & recoveries
- Cumulative token-spend chart
- Per-run detectors & graph route

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Scope of Improvement

Where AgentFuse goes next.

1

Real-time streaming

A live dashboard, not just run replay

2

More detectors

Semantic loops, cost forecasting

3

Fleet view

Supervise many parallel agents at once

4

Adaptive thresholds

Learned drift baselines per task

5

Alerting & policy

Slack / PagerDuty + guardrail rules

6

PyPI package

One-line install: `pip install agentfuse`



Thank You

AgentFuse — keeping autonomous agents on track.

TEAM **Brocode**

github.com/ns-0437/agentfuse

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